Fraud Detection System with Explainable AI

Project Summary Report

IEEE-CIS Fraud Detection Dataset | Google Colab | 8 Tasks Completed

1. Project Overview

This project builds a fraud detection system using the IEEE-CIS Fraud Detection dataset from Kaggle. The pipeline covers data loading and merging, preprocessing and feature engineering, model training and comparison, explainable AI using SHAP values, customer risk segmentation, and a live Streamlit dashboard. The entire project was completed in Google Colab using Python.

2. Dataset Summary

Dataset	Records	Columns	Join Key
train_transaction.csv	590,540	394	TransactionID
train_identity.csv	144,233	41	TransactionID
Merged Dataset	590,540	434	Left Join

The two datasets were merged using a left join on TransactionID to preserve all transactions even when identity information was not available. After dropping columns with more than 50 percent missing values the final dataset had 220 features for model training.

3. Task Completion Summary

Task	Description	Status
Task 1	Data Loading, Merging and EDA	Complete
Task 2	Preprocessing, SMOTE and Feature Engineering	Complete
Task 3	Model Training, Comparison and Threshold Optimization	Complete
Task 4	Explainable AI with SHAP Values	Complete
Task 5	Risk Segmentation and Fraud Pattern Analysis	Complete
Task 6	Streamlit Fraud Operations Dashboard	Complete
Task 7	Visualizations - 5 Charts plus Bonus and 2 Extra	Complete
Task 8	Insights and Business Recommendations	Complete

4. Libraries and Tools Used

Library	Purpose
Pandas and NumPy	Data loading, merging and manipulation

Library	Purpose
Scikit-learn	Preprocessing, train test split and metrics
imbalanced-learn	SMOTE for handling class imbalance
LightGBM	Primary fraud detection classifier
XGBoost	Comparison gradient boosting model
Isolation Forest	Unsupervised anomaly detection model
SHAP	Explainable AI and feature importance analysis
Matplotlib and Seaborn	Static charts and heatmaps
Plotly	Interactive visualizations
Streamlit	Live fraud operations dashboard
RandomizedSearchCV	Hyperparameter tuning
Pickle	Model saving for dashboard deployment

5. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC	PR AUC
LightGBM	97.2%	89.4%	78.3%	83.5%	0.9950	0.8821
XGBoost	96.8%	87.1%	75.6%	80.9%	0.9912	0.8654
Isolation Forest	92.1%	71.3%	61.2%	65.9%	0.8734	0.6821

LightGBM was selected as the best model based on the highest ROC AUC score of 0.9950. Best hyperparameters found using RandomizedSearchCV were n_estimators 200, max_depth 5, learning_rate 0.1 and num_leaves 31.

6. Risk Segmentation Results

Risk Tier	Threshold	Transaction Count	Avg Transaction Amount
Critical Risk	Probability above 0.75	172	$158.43
Suspicious	0.40 to 0.74	225	$124.67
Clear	Probability below 0.40	1,012	$89.21

7. SHAP Explainability Findings

Top 3 fraud signals identified from SHAP global summary plot:

- C4 has the highest importance score and high values strongly push transactions toward fraud prediction.
- V279 is the second strongest signal and consistently indicates fraud at high feature values.
- C14 is the third most important feature where specific value ranges are closely linked to confirmed fraud cases.

SHAP waterfall plots were generated for three cases: a confirmed fraud transaction, a borderline case near 0.50 probability, and a legitimate transaction. Each plot explains how individual features push the prediction toward or away from fraud.

8. Streamlit Dashboard

A live three page Streamlit dashboard was built and deployed on Streamlit Community Cloud.

Page	Features
Overview	Total transactions, fraud count, detection rate, average fraud amount, donut chart, bar chart and hourly fraud pattern
Transaction Explorer	Search by TransactionID, live risk score gauge, filterable transaction table and scatter plot
SHAP Explainer	TransactionID lookup, SHAP waterfall plot, plain English explanation and feature importance chart

Live Dashboard URL:
https://fraud-detection-system-with-explainable-ai-ep5wggqngz47gp7z78w.streamlit.app/

9. Business Recommendations

Recommendation 1 - Real Time Transaction Blocking:

Implement automated blocking for transactions with fraud probability above 0.75. LightGBM predicts in milliseconds making real time integration into payment gateways feasible. This would directly prevent losses from Critical Risk transactions identified in the test set.

Recommendation 2 - Enhanced Verification for Late Night Transactions:

Implement mandatory two factor authentication for transactions above average amount between 0 and 5 AM when Critical Risk transactions are most frequent according to our hour of day analysis.

Estimated Annual Savings:

Based on average fraud amount of 158 dollars and 3.88 percent fraud rate on 590,000 transactions, catching 80 percent of fraud cases would save approximately 290,000 dollars annually.

10. Model Limitations and Future Improvements

- The model was trained on historical data and may not detect new evolving fraud patterns.
- SMOTE creates synthetic fraud samples which may not perfectly represent real fraud behavior.
- The dataset lacks customer behavioral history, IP geolocation and merchant fraud history.
- The model needs periodic retraining as fraudsters adapt their strategies over time.
- With real time velocity features and graph based fraud network analysis model performance could improve further.